

Introduction to SQL

1. What does SQL stand for?

- A) Structured Question Language
- B) Structured Query Language
- C) Simple Query Language
- D) Standard Query Language

Answer: B) Structured Query Language

2. Which organization standardized SQL?

- A) IEEE
- B) ANSI
- C) W3C
- D) ISO

Answer: B) ANSI (American National Standards Institute)

3. Which SQL statement is used to extract data from a database?

- A) GET
- B) OPEN
- C) EXTRACT
- D) SELECT

Answer: D) SELECT

4. Which SQL statement is used to update data in a database?

- A) SAVE
- B) MODIFY
- C) UPDATE
- D) CHANGE

Answer: C) UPDATE

5. Which SQL statement is used to delete data from a database?

- A) REMOVE
- B) DELETE
- C) COLLAPSE
- D) DROP

Answer: B) DELETE

6. Which SQL statement is used to insert new data into a database?

- A) ADD RECORD
- B) ADD NEW
- C) INSERT INTO
- D) INSERT NEW

Answer: C) INSERT INTO

7. With SQL, how do you select a column named "FirstName" from a table named "Persons"?

- A) SELECT FirstName FROM Persons
- B) EXTRACT FirstName FROM Persons
- C) SELECT Persons.FirstName
- D) GET FirstName FROM Persons

Answer: A) SELECT FirstName FROM Persons

8. With SQL, how do you select all the columns from a table named "Persons"?

- A) SELECT [all] FROM Persons
- B) SELECT Persons
- C) SELECT * FROM Persons
- D) SELECT *.Persons

Answer: C) SELECT * FROM Persons

9. With SQL, how do you select all the records from a table named "Persons" where the value of the column "FirstName" is "Peter"?

- A) SELECT * FROM Persons WHERE FirstName='Peter'
- B) SELECT * FROM Persons WHERE FirstName LIKE 'Peter'

C) SELECT [all] FROM Persons WHERE FirstName='Peter'

D) SELECT * FROM Persons WHERE FirstName<>'Peter'

Answer: A) SELECT * FROM Persons WHERE FirstName='Peter'

10. With SQL, how do you select all the records from a table named "Persons" where the value of the column "FirstName" starts with an "a"?

A) SELECT * FROM Persons WHERE FirstName LIKE '%a'

B) SELECT * FROM Persons WHERE FirstName LIKE 'a%'

C) SELECT * FROM Persons WHERE FirstName='%a'

D) SELECT * FROM Persons WHERE FirstName='a'

Answer: B) SELECT * FROM Persons WHERE FirstName LIKE 'a%'

11. The OR operator displays a record if:

A) All conditions listed are true

B) Any of the conditions listed are true

C) None of the conditions listed are true

D) The first condition is true

Answer: B) Any of the conditions listed are true

12. The AND operator displays a record if:

A) All the conditions listed are true

B) Any of the conditions listed are true

C) The first condition is true

D) The last condition is true

Answer: A) All the conditions listed are true

13. Which SQL keyword is used to sort the result-set?

A) SORT BY

B) ORDER

C) ORDER BY

D) SORT

Answer: C) ORDER BY

14. With SQL, how can you return all the records from a table named "Persons" sorted descending by "FirstName"?

- A) SELECT * FROM Persons ORDER BY FirstName DESC
- B) SELECT * FROM Persons SORT BY 'FirstName' DESC
- C) SELECT * FROM Persons ORDER FirstName DESC
- D) SELECT * FROM Persons SORT FirstName DESC

Answer: A) SELECT * FROM Persons ORDER BY FirstName DESC

15. Which SQL statement is used to create a new table in a database?

- A) CREATE TABLE
- B) CREATE DATABASE
- C) CREATE DB
- D) NEW TABLE

Answer: A) CREATE TABLE

16. Which SQL statement is used to create a stored procedure?

- A) CREATE PROCEDURE
- B) NEW PROCEDURE
- C) MAKE PROCEDURE
- D) ADD PROCEDURE

Answer: A) CREATE PROCEDURE

17. How can you change "Hansen" into "Nilsen" in the "LastName" column in the Persons table?

- A) UPDATE Persons SET LastName='Nilsen' WHERE LastName='Hansen'
- B) MODIFY Persons SET LastName='Nilsen' WHERE LastName='Hansen'
- C) UPDATE Persons SET LastName='Hansen' INTO LastName='Nilsen'
- D) CHANGE Persons SET LastName='Nilsen' WHERE LastName='Hansen'

Answer: A) UPDATE Persons SET LastName='Nilsen' WHERE LastName='Hansen'

18. Which SQL statement is used to delete a table?

- A) DELETE TABLE
- B) REMOVE TABLE
- C) DROP TABLE
- D) ERASE TABLE

Answer: C) DROP TABLE

19. Which SQL keyword is used to return only different values?

- A) COUNT
- B) DISTINCT
- C) DIFFERENT
- D) UNIQUE

Answer: B) DISTINCT

20. Which SQL keyword is used to count the number of rows in a table?

- A) SUM
- B) COUNT
- C) NUMBER
- D) ADD

Answer: B) COUNT

21. What is the most common type of Join?

- A) INSIDE JOIN
- B) INNER JOIN
- C) JOINED
- D) LINKED JOIN

Answer: B) INNER JOIN

22. Which operator is used to select values within a range?

- A) WITHIN
- B) RANGE

C) BETWEEN

D) IN

Answer: C) BETWEEN

23. The NOT NULL constraint enforces a column to:

A) Accept NULL values

B) Not accept NULL values

C) Be unique

D) Be a primary key

Answer: B) Not accept NULL values

24. Which constraint uniquely identifies each record in a database table?

A) UNIQUE

B) PRIMARY KEY

C) FOREIGN KEY

D) CHECK

Answer: B) PRIMARY KEY

25. Which constraint is used to ensure that all values in a column are different?

A) UNIQUE

B) DEFAULT

C) CHECK

D) INDEX

Answer: A) UNIQUE

26. SQL is a:

A) Procedural Language

B) Non-Procedural Language

C) Low-level Language

D) Object-Oriented Language

Answer: B) Non-Procedural Language

27. Which subset of SQL is used to define the database structure?

- A) DML
- B) DDL
- C) DCL
- D) TCL

Answer: B) DDL (Data Definition Language)

28. Which subset of SQL is used to manipulate data?

- A) DML
- B) DDL
- C) DCL
- D) TCL

Answer: A) DML (Data Manipulation Language)

29. Which subset of SQL includes commands like GRANT and REVOKE?

- A) DML
- B) DDL
- C) DCL
- D) TCL

Answer: C) DCL (Data Control Language)

30. Which subset of SQL includes commands like COMMIT and ROLLBACK?

- A) DML
- B) DDL
- C) DCL
- D) TCL

Answer: D) TCL (Transaction Control Language)

31. The command `TRUNCATE TABLE` belongs to which category?

- A) DML

B) DDL

C) DCL

D) TCL

Answer: B) DDL (in most standard definitions, though it acts on data, it resets structure)

32. Which character is used to represent any single character in a LIKE pattern?

A) %

B) *

C) _ (Underscore)

D) ?

Answer: C) _ (Underscore)

33. Which character is used to represent zero or more characters in a LIKE pattern?

A) %

B) *

C) _

D) #

Answer: A) %

34. What is a Foreign Key?

A) A key that is foreign to the database

B) A primary key of one table that appears as an attribute in another table

C) A unique key

D) A key that allows nulls

Answer: B) A primary key of one table that appears as an attribute in another table

35. The `HAVING` clause is used in combination with:

A) WHERE

B) GROUP BY

C) ORDER BY

D) SELECT

Answer: B) GROUP BY

36. To remove duplicate rows from the result set of a SELECT use:

A) NO DUPLICATE

B) UNIQUE

C) DISTINCT

D) SINGLE

Answer: C) DISTINCT

37. Which function returns the smallest value of the selected column?

A) LOW()

B) MIN()

C) SMALL()

D) LEAST()

Answer: B) MIN()

38. Which function returns the largest value of the selected column?

A) HIGH()

B) MAX()

C) LARGE()

D) GREATEST()

Answer: B) MAX()

39. Which function returns the average value of a numeric column?

A) AVERAGE()

B) AVG()

C) MEAN()

D) MEDIAN()

Answer: B) AVG()

40. Which function returns the sum of a numeric column?

- A) ADD()
- B) TOTAL()
- C) SUM()
- D) COUNT()

Answer: C) SUM()

41. `SELECT \* FROM Customers WHERE City IN ('Paris','London');` selects:

- A) Customers from Paris only
- B) Customers from London only
- C) Customers from Paris and London
- D) Customers not from Paris or London

Answer: C) Customers from Paris and London

42. What does the `AS` keyword do?

- A) Renames a table permanently
- B) Gives an alias to a column or table
- C) Copies a table
- D) Deletes a column

Answer: B) Gives an alias to a column or table

43. Which join returns all records from the left table, and the matched records from the right table?

- A) RIGHT JOIN
- B) LEFT JOIN
- C) INNER JOIN
- D) FULL JOIN

Answer: B) LEFT JOIN

44. Which join returns all records when there is a match in either left or right table?

- A) RIGHT JOIN
- B) LEFT JOIN
- C) INNER JOIN

D) FULL JOIN

Answer: D) FULL JOIN

45. The UNION operator selects only:

A) Distinct values

B) Duplicate values

C) Null values

D) All values including duplicates (unless UNION ALL is used)

Answer: A) Distinct values

46. Which operator is used to combine the result-set of two or more SELECT statements?

A) JOIN

B) MERGE

C) UNION

D) COMBINE

Answer: C) UNION

47. The `CREATE DATABASE` statement is used to:

A) Create a new SQL table

B) Create a new SQL database

C) Create a new trigger

D) Create a new index

Answer: B) Create a new SQL database

48. The `ALTER TABLE` statement is used to:

A) Add, delete, or modify columns in an existing table

B) Create a new table

C) Delete a table

D) Insert data into a table

Answer: A) Add, delete, or modify columns in an existing table

49. To delete a specific column from a table, which syntax is correct?

- A) ALTER TABLE table_name DROP COLUMN column_name
- B) DELETE COLUMN column_name FROM table_name
- C) DROP COLUMN column_name FROM table_name
- D) REMOVE COLUMN column_name

Answer: A) ALTER TABLE table_name DROP COLUMN column_name

50. The `DEFAULT` constraint is used to:

- A) Enforce unique values
- B) Prevent NULL values
- C) Provide a default value for a column when none is specified
- D) Index a column

Answer: C) Provide a default value for a column when none is specified

51. An index:

- A) Slows down queries
- B) Speeds up data retrieval
- C) Is used for data backup
- D) Is a type of table

Answer: B) Speeds up data retrieval

52. `CREATE INDEX` statement is used to:

- A) Create a new table
- B) Create indexes in tables
- C) Create a database
- D) Create a primary key

Answer: B) Create indexes in tables

53. Auto Increment allows a unique number to be generated:

- A) Automatically when a new record is inserted
- B) Manually by the user

C) Only for Foreign Keys

D) Randomly

Answer: A) Automatically when a new record is inserted

54. A `VIEW` in SQL is:

A) A physical table

B) A virtual table based on the result-set of an SQL statement

C) A backup file

D) A stored procedure

Answer: B) A virtual table based on the result-set of an SQL statement

55. Which SQL function returns the current date and time?

A) GETDATE() / NOW()

B) TIME()

C) TODAY()

D) CURRENT()

Answer: A) GETDATE() / NOW() (Depending on DBMS)

56. The `IS NULL` operator is used to test for:

A) Zero values

B) Empty strings

C) Empty (NULL) values

D) False values

Answer: C) Empty (NULL) values

57. The `IS NOT NULL` operator is used to test for:

A) Non-empty values

B) Zero values

C) True values

D) Valid keys

Answer: A) Non-empty values

58. What is the difference between `DELETE` and `TRUNCATE`?

- A) DELETE is DDL, TRUNCATE is DML
- B) DELETE can have a WHERE clause, TRUNCATE cannot
- C) TRUNCATE is slower than DELETE
- D) They are exactly the same

Answer: B) DELETE can have a WHERE clause, TRUNCATE cannot

59. SQL comments can be written using:

- A) //
- B) -- (Double dash)
- C) #
- D) ;

Answer: B) -- (Double dash)

60. Which command undoes transactions that have not been saved to the database?

- A) SAVE
- B) COMMIT
- C) ROLLBACK
- D) RESTORE

Answer: C) ROLLBACK

61. `SELECT TOP` clause is used to:

- A) Select the highest value
- B) Select a column named TOP
- C) Specify the number of records to return
- D) Select the first row only

Answer: C) Specify the number of records to return

62. To select records starting with any character, followed by 'L', followed by any characters:

- A) LIKE 'L%'

B) LIKE '_L%'

C) LIKE '%L'

D) LIKE 'L_'

Answer: B) LIKE '_L%'

63. Which constraint limits the value range that can be placed in a column?

A) RANGE

B) CHECK

C) DEFAULT

D) LIMIT

Answer: B) CHECK

64. `DROP DATABASE` command:

A) Deletes the database schema

B) Deletes the database and all its data

C) Deletes tables but keeps the database

D) Clears data from tables

Answer: B) Deletes the database and all its data

65. The `GROUP BY` statement groups rows that have the same values into:

A) Summary rows

B) Distinct rows

C) Sorted rows

D) Filtered rows

Answer: A) Summary rows

66. What is the Cartesian Product in SQL?

A) The result of an INNER JOIN

B) The result of a LEFT JOIN

C) The result of joining every row of one table to every row of another table (CROSS JOIN)

D) A mathematical error

Answer: C) The result of joining every row of one table to every row of another table (CROSS JOIN)

67. Which SQL keyword allows you to combine the results of two queries?

- A) COMBINE
- B) UNION
- C) JOIN
- D) MERGE

Answer: B) UNION

68. The `EXISTS` operator is used to:

- A) Check if a table exists
- B) Check if a record exists in a subquery
- C) Check if a column exists
- D) Check database status

Answer: B) Check if a record exists in a subquery

69. The `ANY` and `ALL` operators are used with:

- A) A WHERE or HAVING clause
- B) A CREATE statement
- C) An INSERT statement
- D) A DROP statement

Answer: A) A WHERE or HAVING clause

70. `SELECT INTO` statement:

- A) Selects data into a variable
- B) Copies data from one table into a new table
- C) Inserts data into an existing table
- D) Updates data

Answer: B) Copies data from one table into a new table

71. `INSERT INTO SELECT` statement:

- A) Creates a new table
- B) Copies data from one table and inserts it into an existing table
- C) Selects data to view
- D) Deletes data

Answer: B) Copies data from one table and inserts it into an existing table

72. `CASE` statement in SQL:

- A) Creates a new case
- B) Goes through conditions and returns a value when the first condition is met
- C) Checks for errors
- D) Is not valid in SQL

Answer: B) Goes through conditions and returns a value when the first condition is met

73. `NULL` functions like `IFNULL()`, `ISNULL()`, `COALESCE()` are used to:

- A) Create NULL values
- B) Return an alternative value if an expression is NULL
- C) Delete NULL values
- D) Find NULL values

Answer: B) Return an alternative value if an expression is NULL

74. A self-join is:

- A) A regular join but the table is joined with itself
- B) Joining a table with a view
- C) Joining a table with a backup
- D) Joining without a condition

Answer: A) A regular join but the table is joined with itself

75. In `SELECT \* FROM A, B`, what kind of join is performed?

- A) Inner Join
- B) Outer Join
- C) Cross Join

D) Self Join

Answer: C) Cross Join

76. Which normal form ensures that there are no repeating groups?

A) 1NF

B) 2NF

C) 3NF

D) BCNF

Answer: A) 1NF

77. Which statement gives users access privileges to the database?

A) ALLOW

B) GRANT

C) PERMIT

D) ACCESS

Answer: B) GRANT

78. Which statement withdraws access privileges given with the GRANT statement?

A) REMOVE

B) DELETE

C) REVOKE

D) DENY

Answer: C) REVOKE

79. A composite key is:

A) A foreign key

B) A primary key consisting of multiple columns

C) A candidate key

D) A super key

Answer: B) A primary key consisting of multiple columns

80. What is a Candidate Key?

- A) A key that references another table
- B) A set of attributes that can uniquely identify a tuple
- C) The selected primary key
- D) A non-unique key

Answer: B) A set of attributes that can uniquely identify a tuple

81. RDBMS stands for:

- A) Relational Data Base Management System
- B) Regional Data Base Management System
- C) Real Data Base Management System
- D) Rapid Data Base Management System

Answer: A) Relational Data Base Management System

82. Which of the following is NOT a logical operator?

- A) AND
- B) OR
- C) NOT
- D) IF

Answer: D) IF (IF is a control flow statement, not a logical operator in WHERE clauses like AND/OR/NOT)

83. The `%` wildcard matches:

- A) One character
- B) Zero or one character
- C) Zero or more characters
- D) One or more characters

Answer: C) Zero or more characters

84. SQL `aliases` are used to:

- A) Give a table or a column a temporary name

- B) Encrypt column names
- C) Delete columns
- D) Permanently rename columns

Answer: A) Give a table or a column a temporary name

85. Which statement is used to add a column to an existing table?

- A) INSERT COLUMN
- B) MODIFY TABLE ADD
- C) ALTER TABLE ADD
- D) UPDATE TABLE ADD

Answer: C) ALTER TABLE ADD

86. Which statement is used to delete a column from an existing table?

- A) DELETE COLUMN
- B) DROP COLUMN
- C) ALTER TABLE DROP COLUMN
- D) REMOVE COLUMN

Answer: C) ALTER TABLE DROP COLUMN

87. Which data type is used to store text of variable length?

- A) CHAR
- B) VARCHAR
- C) TEXT
- D) BLOB

Answer: B) VARCHAR

88. Which data type is typically used for whole numbers?

- A) FLOAT
- B) DECIMAL
- C) INT
- D) REAL

Answer: C) INT

89. To combine column values in a result set (concatenation), some SQL dialects use:

A) &

B) + or ||

C) *

D) #

Answer: B) + or || (CONCAT function also used)

90. `LEN()` or `LENGTH()` function returns:

A) The value of the field

B) The length of a text field

C) The size of the table

D) The number of rows

Answer: B) The length of a text field

91. `UCASE()` or `UPPER()` converts a field to:

A) Lower case

B) Upper case

C) Title case

D) Camel case

Answer: B) Upper case

92. `LCASE()` or `LOWER()` converts a field to:

A) Lower case

B) Upper case

C) Text format

D) Number format

Answer: A) Lower case

93. `ROUND()` function allows you to:

- A) Round a numeric field to the number of decimals specified
- B) Round a date
- C) Create a circle
- D) Loop through data

Answer: A) Round a numeric field to the number of decimals specified

94. Which of the following is an Aggregate Function?

- A) ABS()
- B) CEILING()
- C) FLOOR()
- D) COUNT()

Answer: D) COUNT()

95. In `SELECT count(ProductID) FROM Products`, what is returned?

- A) The sum of ProductIDs
- B) The number of products
- C) The average ProductID
- D) The unique ProductIDs

Answer: B) The number of products

96. Which operator returns true if the operand is within a range of values?

- A) IN
- B) BETWEEN
- C) LIKE
- D) EXISTS

Answer: B) BETWEEN

97. Which operator returns true if the operand matches a pattern?

- A) IN
- B) BETWEEN
- C) LIKE

D) ANY

Answer: C) LIKE

98. Which operator returns true if the operand is equal to one of a list of expressions?

A) IN

B) BETWEEN

C) LIKE

D) ALL

Answer: A) IN

99. The `HAVING` clause was added to SQL because:

A) The WHERE keyword could not be used with aggregate functions

B) The SELECT keyword was insufficient

C) The ORDER BY clause needed support

D) It allows faster querying

Answer: A) The WHERE keyword could not be used with aggregate functions

100. When using `ORDER BY`, the default sort order is:

A) Descending

B) Random

C) Ascending

D) Reversed

Answer: C) Ascending